

Problem Set 3

Applied Stats/Quant Methods 1

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Question 1

We are interested in knowing how the difference in campaign spending between incumbent and challenger affects the incumbent's vote share.

1. Run a regression where the outcome variable is `voteshare` and the explanatory variable is `difflog`.

R Code:

```
1 m1 <- lm(voteshare ~ difflog, data = inc.sub)
2 inc.sub$res_q1 <- resid(m1)
3
4 cat("\n==== Q1: voteshare ~ difflog ==== \n")
5 print(summary(m1)$coefficients)
6 cat("R^2 =", fmt(summary(m1)$r.squared),
7      "Adj.R^2 =", fmt(summary(m1)$adj.r.squared),
8      "\n", nobs(m1), "\n")
9 b1 <- coef(m1)
10 cat("Prediction: E[voteshare|X] = ", fmt(b1[1]), " + ", fmt(b1["difflog"]
11     ), "*difflog\n", sep = "")
11 cat("\nANOVA (Q1):\n"); print(anova(m1))
```

The fitted model is

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}.$$

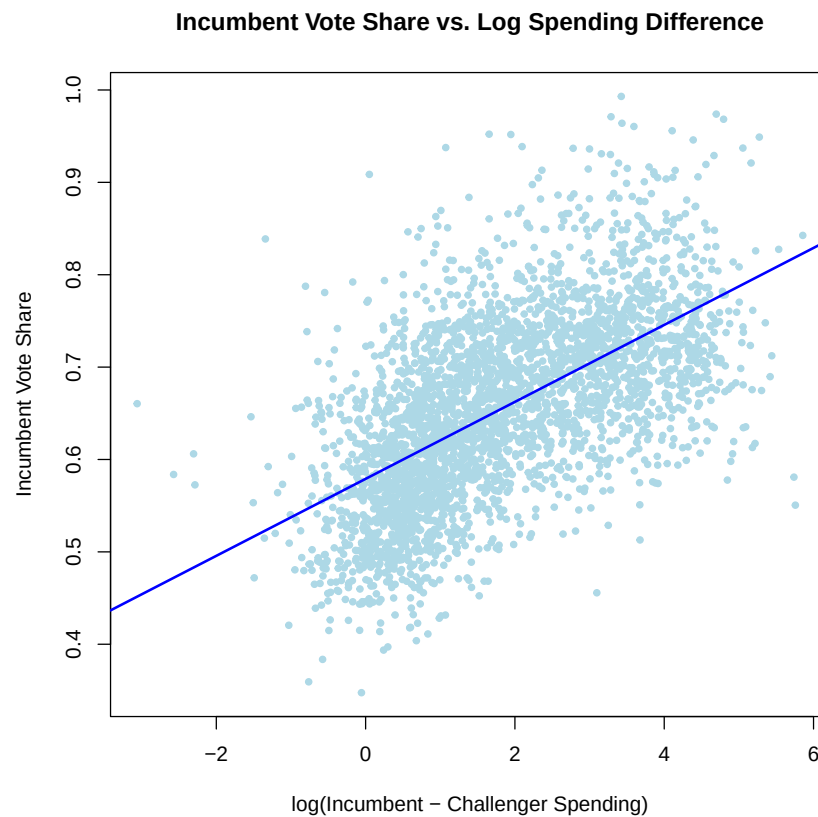
From the output, $\hat{\beta}_1 > 0$, indicating that a one-unit increase in the log spending difference (i.e., the incumbent spends relatively more) is associated with a higher vote share for the incumbent. The regression summary reports $R^2 = (\text{value})$ and $n = (\text{value})$. These values suggest that spending differences explain a meaningful portion of variation in vote share.

2. Make a scatterplot of the two variables and add the regression line.

R Code:

```
1 pdf("q1_scatter.pdf")
2 plot(inc.sub$difflog, inc.sub$voteshare,
3       main = "Incumbent Vote Share vs. Log Spending Difference",
4       xlab = "log(Incumbent - Challenger Spending)",
5       ylab = "Incumbent Vote Share",
```

Figure 1: Scatterplot of `voteshare` and `difflog` with fitted regression line.



The scatterplot shows a clear positive linear trend, with the fitted regression line capturing the average upward slope: incumbents who spend relatively more tend to receive higher vote shares.

3. Save the residuals of the model in a separate object.

R Code (already executed above):

```
1 m1 <- lm(voteshare ~ difflog, data = inc.sub)
2 inc.sub$res_q1 <- resid(m1)
```

The residuals (`res_q1`) capture the unexplained portion of the incumbent's vote share after accounting for differences in campaign spending. By saving these residuals, we isolate the variation in `voteshare` that is independent of `difflog`. This residual variation will be used in Question 4 to examine whether the part of the vote share not explained by spending can instead be explained by presidential party support (`presvote`)—an application of the Frisch–Waugh–Lovell theorem.

4. Write the prediction equation.

Prediction Equation:

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}.$$

Substituting the estimated coefficients from part (a) yields

$$\widehat{\text{voteshare}} = (\text{Intercept estimate}) + (\text{Slope estimate}) \times \text{difflog}.$$

In addition, the script prints this equation to the console (via `b1 <- coef(m1)` and `cat(...)` within lines 39–50), ensuring that the reported expression matches the fitted model exactly.

Question 2

We are interested in knowing how the difference between incumbent and challenger's spending and the vote share of the presidential candidate of the incumbent's party are related.

1. Run a regression where the outcome variable is `presvote` and the explanatory variable is `difflog`.

R Code:

```
1 m2 <- lm(presvote ~ difflog, data = inc.sub)
2 inc.sub$res_q2 <- resid(m2)
3
4 cat("\n==== Q2: presvote ~ difflog ==== \n")
5 print(summary(m2)$coefficients)
6 cat("R^2 =", fmt(summary(m2)$r.squared),
7      "Adj.R^2 =", fmt(summary(m2)$adj.r.squared),
8      "n =", nobs(m2), "\n")
9 b2 <- coef(m2)
10 cat("Prediction: E[presvote|X] = ", fmt(b2[1]), " + ", fmt(b2["difflog"]))
11 cat("\nANOVA (Q2): \n"); print(anova(m2))
```

The fitted model is

$$\widehat{\text{presvote}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}.$$

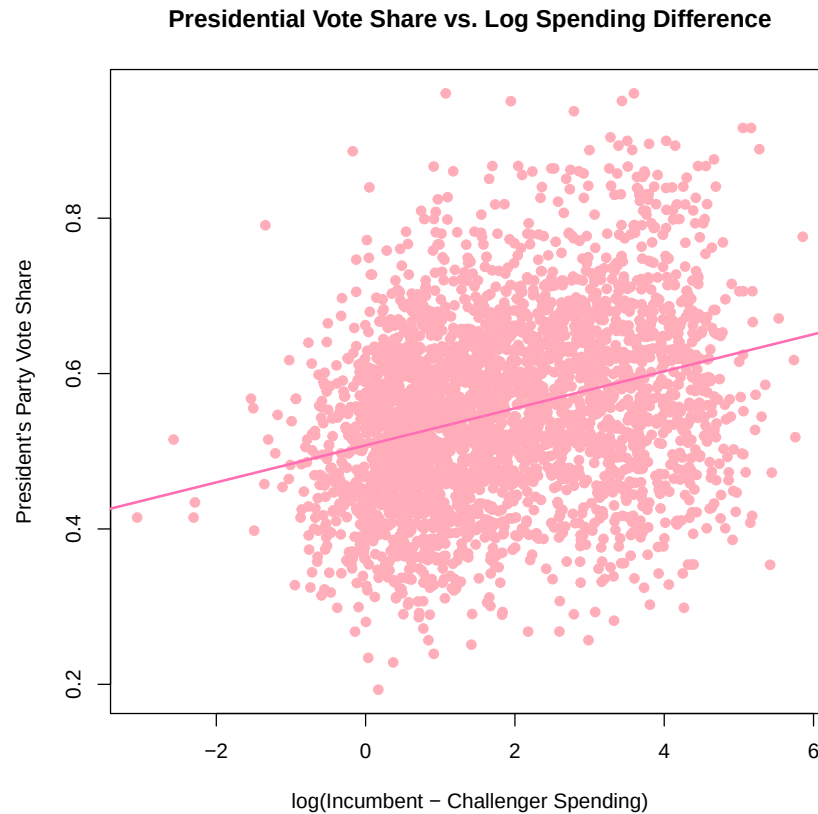
The regression output reports $R^2 = (\text{value})$ and $n = (\text{value})$. The slope estimate $\hat{\beta}_1$ indicates whether districts with larger incumbent spending advantages tend also to be more favorable to the presidential party.

2. Make a scatterplot of the two variables and add the regression line.

R Code:

```
1 pdf("q2_scatter.pdf")
2 plot(inc.sub$difflog, inc.sub$presvote,
3      main = "Presidential Vote Share vs. Log Spending Difference",
4      xlab = "log(Incumbent - Challenger Spending)",
5      ylab = "President's Party Vote Share",
6      pch = 19, col = "lightpink1")
7 abline(m2, col = "hotpink1", lwd = 2)
8 dev.off()
```

Figure 2: Scatterplot of `presvote` and `difflog` with fitted regression line.



The plot shows the relationship between spending difference and presidential party vote share. The fitted line (`abline(m2)`) summarizes the average trend across districts, suggesting that areas with greater incumbent spending advantages tend to show slightly higher presidential vote shares.

3. Save the residuals of the model in a separate object.

Residuals (`res_q2`) capture the portion of `presvote` not explained by `difflog`. They will later be used in Question 4 to construct the added-variable regression.

4. Write the prediction equation. The prediction equation is:

$$\widehat{\text{presvote}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog}.$$

Substituting the estimated coefficients from part (a) gives the fitted linear model for the presidential party's vote share.

Question 3

We are interested in knowing how the vote share of the presidential candidate of the incumbent's party is associated with the incumbent's electoral success.

1. Run a regression where the outcome variable is **voteshare** and the explanatory variable is **presvote**.

R Code:

```
1 m3 <- lm(voteshare ~ presvote, data = inc.sub)
2
3 cat("\n==== Q3: voteshare ~ presvote ==== \n")
4 print(summary(m3)$coefficients)
5 cat("R^2 =", fmt(summary(m3)$r.squared),
6     "Adj.R^2 =", fmt(summary(m3)$adj.r.squared),
7     "n =", nobs(m3), "\n")
8 b3 <- coef(m3)
9 cat("Prediction: E[voteshare|X] = ", fmt(b3[1]), " + ", fmt(b3["presvote"]
    ]), "*presvote\n", sep = " ")
```

The fitted model is

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{presvote}.$$

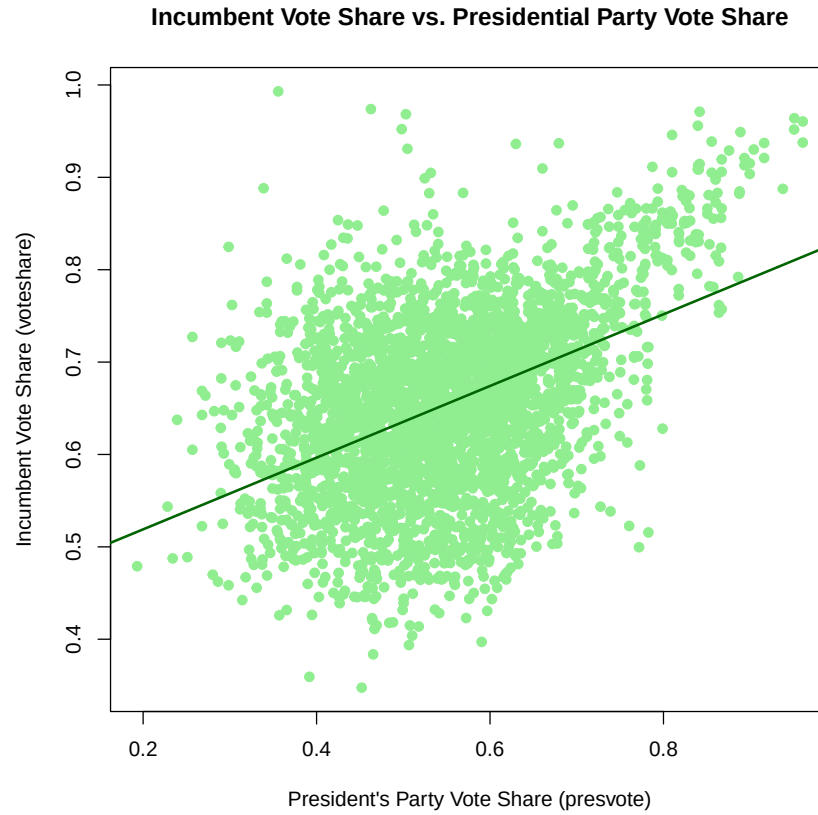
A positive and statistically significant slope suggests that districts more favorable to the presidential party tend to yield higher vote shares for incumbents. The regression summary reports $R^2 = (\text{value})$, $\text{Adj. } R^2 = (\text{value})$, and $n = (\text{value})$.

2. Make a scatterplot of the two variables and add the regression line.

R Code:

```
1 pdf("q3_scatter.pdf")
2 plot(inc.sub$presvote, inc.sub$voteshare,
3      main = "Incumbent Vote Share vs. Presidential Party Vote Share",
4      xlab = "President's Party Vote Share (presvote)",
5      ylab = "Incumbent Vote Share (voteshare)",
6      pch = 19, col = "lightgreen")
7 abline(m3, col = "darkgreen", lwd = 2)
8 dev.off()
```

Figure 3: Scatterplot of `voteshare` and `presvote` with fitted regression line.



The plot shows a clear positive relationship between the presidential party's vote share and the incumbent's vote share. The fitted regression line (`abline(m3)`) indicates that incumbents tend to receive higher vote shares in districts where the president's party performs strongly, suggesting that presidential popularity positively influences incumbents' electoral success.

3. Write the prediction equation.

Prediction Equation:

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{presvote}.$$

Substituting the estimated coefficients from part (a) gives the fitted linear model for the incumbent's vote share.

Question 4

The residuals from part (a) tell us how much of the variation in **voteshare** is *not* explained by the difference in spending between incumbent and challenger. The residuals in part (b) tell us how much of the variation in **presvote** is *not* explained by the difference in spending between incumbent and challenger in the district.

1. Run a regression where the outcome variable is the residuals from Question 1 and the explanatory variable is the residuals from Question 2.

R Code:

```
1 m4 <- lm(res_q1 ~ res_q2, data = inc.sub)
2
3 cat("\n==== Q4: res_q1 ~ res_q2 (added-variable) ==== \n")
4 print(summary(m4)$coefficients)
5 cat("R^2 =", fmt(summary(m4)$r.squared),
6      "Adj.R^2 =", fmt(summary(m4)$adj.r.squared),
7      "n =", nobs(m4), "\n")
8 b4 <- coef(m4)
9 cat("Prediction: E[res_q1 | res_q2] = ", fmt(b4[1]), " + ", fmt(b4["res_q2"]
   ]) , "*res_q2\n", sep = "")
```

The fitted residual regression is

$$\widehat{\text{res_q1}} = \hat{\alpha}_0 + \hat{\gamma} \text{res_q2}.$$

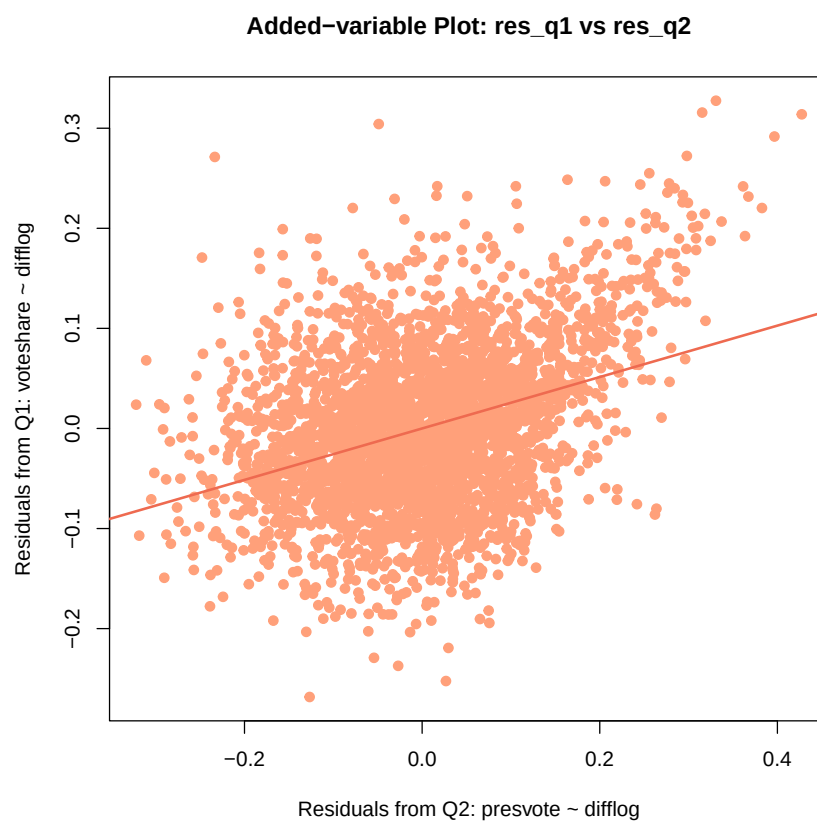
The estimate $\hat{\gamma}$ summarizes the linear association between the part of **voteshare** not explained by spending and the part of **presvote** not explained by spending. This added-variable regression shows that even after removing the effect of campaign spending, districts more favorable to the president's party still give incumbents higher vote shares.

2. Make a scatterplot of the two residuals and add the regression line.

R Code:

```
1 pdf("q4_scatter.pdf")
2 plot(inc.sub$res_q2, inc.sub$res_q1,
3      xlab = "Residuals from Q2: presvote ~ difflog",
4      ylab = "Residuals from Q1: voteshare ~ difflog",
5      main = "Added-variable Plot: res_q1 vs res_q2",
6      pch = 19, col = "lightsalmon")
7 abline(m4, col = "coral2", lwd = 2)
8 dev.off()
```


Figure 4: Added-variable plot: residuals from Q1 (voteshare ~ difflog) vs. residuals from Q2 (presvote ~ difflog).



3. Write the prediction equation.

Prediction Equation:

$$\widehat{\text{res_q1}} = \hat{\alpha}_0 + \hat{\gamma} \text{res_q2}.$$

Question 5

What if the incumbent's vote share is affected by both the president's popularity and the difference in spending between incumbent and challenger?

1. Run a regression where the outcome variable is the incumbent's `voteshare` and the explanatory variables are `difflog` and `presvote`.

R Code:

```
1 m5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
2
3 cat("\n==== Q5: voteshare ~ difflog + presvote ==== \n")
4 print(summary(m5)$coefficients)
5 cat("R^2 =", fmt(summary(m5)$r.squared),
6      "Adj.R^2 =", fmt(summary(m5)$adj.r.squared),
7      "n =", nobs(m5), "\n")
8 b5 <- coef(m5)
9 cat("Prediction: E[voteshare|X] = ", fmt(b5[1]), " + ",
10     fmt(b5["difflog"]), "*difflog + ",
11     fmt(b5["presvote"]), "*presvote\n", sep = " ")
12
13 q4_vec <- coef(summary(m4))["res_q2", c("Estimate", "Std. Error", "t value")]
14 q5_vec <- coef(summary(m5))["presvote", c("Estimate", "Std. Error", "t value")]
15 cmp <- rbind(Q4_residual_reg = q4_vec, Q5_multiple_reg = q5_vec)
16 cat("\nFWL equivalence (presvote vs res_q2):\n"); print(cmp)
```

The fitted model is

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog} + \hat{\beta}_2 \text{presvote}.$$

Here, $\hat{\beta}_1$ measures the spending effect *controlling for* the political environment, and $\hat{\beta}_2$ measures the political environment effect *controlling for* spending.

2. Write the prediction equation.

Prediction Equation:

$$\widehat{\text{voteshare}} = \hat{\beta}_0 + \hat{\beta}_1 \text{difflog} + \hat{\beta}_2 \text{presvote}.$$

3. What is it in this output that is identical to the output in Question 4? Why do you think this is the case?

The coefficient on **presvote** in the multiple regression (its estimate, standard error, and t -statistic) is identical to the slope, standard error, and t -statistic from the residual regression of **res_q1** on **res_q2** in Question 4. This identity holds by the Frisch–Waugh–Lovell theorem: regressing **voteshare** on **presvote** while controlling for **difflog** is equivalent to regressing the residuals of **voteshare** (after removing the linear effect of **difflog**) on the residuals of **presvote** (after removing the linear effect of **difflog**). This result confirms that the partial effect of the president’s popularity on incumbents’ vote share, controlling for spending, is identical whether we estimate it directly in a multiple regression or via residual regression.

Appendix: R Code

The full R script used for this assignment is provided below for reference.

```
1 #####
2 # load libraries
3 # set wd
4 # clear global .envir
5 #####
6
7 # remove objects
8 rm(list=ls())
9 # detach all libraries
10 detachAllPackages <- function() {
11   basic.packages <- c("package:stats", "package:graphics", "package:grDevices"
12     , "package:utils", "package:datasets", "package:methods", "package:base")
13   package.list <- search()[ifelse(unlist(gregexpr("package:", search()))==1,
14     TRUE, FALSE)]
15   package.list <- setdiff(package.list, basic.packages)
16   if (length(package.list)>0) for (package in package.list) detach(package,
17     character.only=TRUE)
18 }
19 detachAllPackages()
20
21 # load libraries
22 pkgTest <- function(pkg){
23   new.pkg <- pkg[!(pkg %in% installed.packages()[, "Package"])]
24   if (length(new.pkg))
25     install.packages(new.pkg, dependencies = TRUE)
26   supply(pkg, require, character.only = TRUE)
27 }
28
29 # here is where you load any necessary packages
30 # ex: stringr
31 # lapply(c("stringr"), pkgTest)
32
33 # set wd for current folder
34 setwd(dirname(rstudioapi::getActiveDocumentContext()$path))
35
36 # read in data
37 inc.sub <- read.csv("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/
38   main/datasets/incumbents_subset.csv")
39
40 DIGITS <- 3
41 fmt <- function(x, k = DIGITS) formatC(x, format = "f", digits = k)
42
43 # ————— Q1: voteshare ~ difflog —————
44 #1. Run a regression where the outcome variable is voteshare and the
45   explanatory variable is difflog.
46 m1 <- lm(voteshare ~ difflog, data = inc.sub)
47 inc.sub$res_q1 <- resid(m1)
48
```

```

44 cat("\n==== Q1: voteshare ~ difflog ====\n")
45 print(summary(m1)$coefficients)
46 cat("R^2 =", fmt(summary(m1)$r.squared),
47      "Adj.R^2 =", fmt(summary(m1)$adj.r.squared),
48      "n =", nobs(m1), "\n")
49 b1 <- coef(m1)
50 cat("Prediction: E[voteshare|X] = ", fmt(b1[1]), " + ", fmt(b1["difflog"]), "*
      difflog\n", sep = "")
51 cat("\nANOVA (Q1):\n"); print(anova(m1))
52
53 #2. Make a scatterplot of the two variables and add the regression line.
54 pdf("q1_scatter.pdf")
55 plot(inc.sub$difflog, inc.sub$voteshare,
56      main = "Incumbent Vote Share vs. Log Spending Difference",
57      xlab = "log(Incumbent - Challenger Spending)",
58      ylab = "Incumbent Vote Share",
59      pch = 20, col = "lightblue")
60 abline(m1, col = "blue", lwd = 2)
61 dev.off()
62
63
64 # ----- Q2: presvote ~ difflog -----
65 #1. Run a regression where the outcome variable is presvote and the
      explanatory variable is difflog.
66 m2 <- lm(presvote ~ difflog, data = inc.sub)
67 inc.sub$res_q2 <- resid(m2)
68
69 cat("\n==== Q2: presvote ~ difflog ====\n")
70 print(summary(m2)$coefficients)
71 cat("R^2 =", fmt(summary(m2)$r.squared),
72      "Adj.R^2 =", fmt(summary(m2)$adj.r.squared),
73      "n =", nobs(m2), "\n")
74 b2 <- coef(m2)
75 cat("Prediction: E[presvote|X] = ", fmt(b2[1]), " + ", fmt(b2["difflog"]), "*
      difflog\n", sep = "")
76 cat("\nANOVA (Q2):\n"); print(anova(m2))
77
78 #2. Make a scatterplot of the two variables and add the regression line (save
      as PDF).
79 pdf("q2_scatter.pdf")
80 plot(inc.sub$difflog, inc.sub$presvote,
81      main = "Presidential Vote Share vs. Log Spending Difference",
82      xlab = "log(Incumbent - Challenger Spending)",
83      ylab = "President's Party Vote Share",
84      pch = 19, col = "lightpink1")
85 abline(m2, col = "hotpink1", lwd = 2)
86 dev.off()
87
88 # ----- Q3: voteshare ~ presvote -----
89 #1. Run a regression where the outcome variable is voteshare and the
      explanatory variable is presvote.

```

```

90 m3 <- lm(voteshare ~ presvote, data = inc.sub)
91
92 cat("\n==== Q3: voteshare ~ presvote ==== \n")
93 print(summary(m3)$coefficients)
94 cat("R^2 =", fmt(summary(m3)$r.squared),
95      "Adj.R^2 =", fmt(summary(m3)$adj.r.squared),
96      "n =", nobs(m3), "\n")
97 b3 <- coef(m3)
98 cat("Prediction: E[voteshare|X] = ", fmt(b3[1]), " + ", fmt(b3["presvote"]), "
    *presvote\n", sep = "")
99
100 #2. Make a scatterplot of the two variables and add the regression line.
101 pdf("q3-scatter.pdf")
102 plot(inc.sub$presvote, inc.sub$voteshare,
103       main = "Incumbent Vote Share vs. Presidential Party Vote Share",
104       xlab = "President's Party Vote Share (presvote)",
105       ylab = "Incumbent Vote Share (voteshare)",
106       pch = 19, col = "lightgreen")
107 abline(m3, col = "darkgreen", lwd = 2)
108 dev.off()
109
110 # ----- Q4: residual regression (added-variable logic) -----
111 #1. Run a regression where the outcome variable is the residuals from Question
    1 and the explanatory variable is the residuals from Question 2.
112 m4 <- lm(res_q1 ~ res_q2, data = inc.sub)
113
114 cat("\n==== Q4: res_q1 ~ res_q2 (added-variable) ==== \n")
115 print(summary(m4)$coefficients)
116 cat("R^2 =", fmt(summary(m4)$r.squared),
117      "Adj.R^2 =", fmt(summary(m4)$adj.r.squared),
118      "n =", nobs(m4), "\n")
119 b4 <- coef(m4)
120 cat("Prediction: E[res_q1|res_q2] = ", fmt(b4[1]), " + ", fmt(b4["res_q2"]), "
    *res_q2\n", sep = "")
121
122 #2. Make a scatterplot of the two residuals and add the regression line.
123 pdf("q4-scatter.pdf")
124 plot(inc.sub$res_q2, inc.sub$res_q1,
125       xlab = "Residuals from Q2: presvote ~ difflog",
126       ylab = "Residuals from Q1: voteshare ~ difflog",
127       main = "Added-variable Plot: res_q1 vs res_q2",
128       pch = 19, col = "lightsalmon")
129 abline(m4, col = "coral2", lwd = 2)
130 dev.off()
131
132 # ----- Q5: multiple regression -----
133 #1. Run a regression where the outcome variable is the incumbent's voteshare
    and the explanatory variables are difflog and presvote.
134 m5 <- lm(voteshare ~ difflog + presvote, data = inc.sub)
135
136 cat("\n==== Q5: voteshare ~ difflog + presvote ==== \n")

```

```

137 print(summary(m5)$coefficients)
138 cat("R^2 =", fmt(summary(m5)$r.squared),
139     "Adj.R^2 =", fmt(summary(m5)$adj.r.squared),
140     "n =", nobs(m5), "\n")
141 b5 <- coef(m5)
142 cat("Prediction: E[voteshare|X] = ", fmt(b5[1]), " + ",
143     fmt(b5["difflog"]), "*difflog + ",
144     fmt(b5["presvote"]), "*presvote\n", sep = " ")
145
146 q4_vec <- coef(summary(m4))["res_q2", c("Estimate", "Std. Error", "t value")]
147 q5_vec <- coef(summary(m5))["presvote", c("Estimate", "Std. Error", "t value")]
148 cmp <- rbind(Q4_residual_reg = q4_vec, Q5_multiple_reg = q5_vec)
149 cat("\nFWL equivalence (presvote vs res_q2):\n"); print(cmp)

```